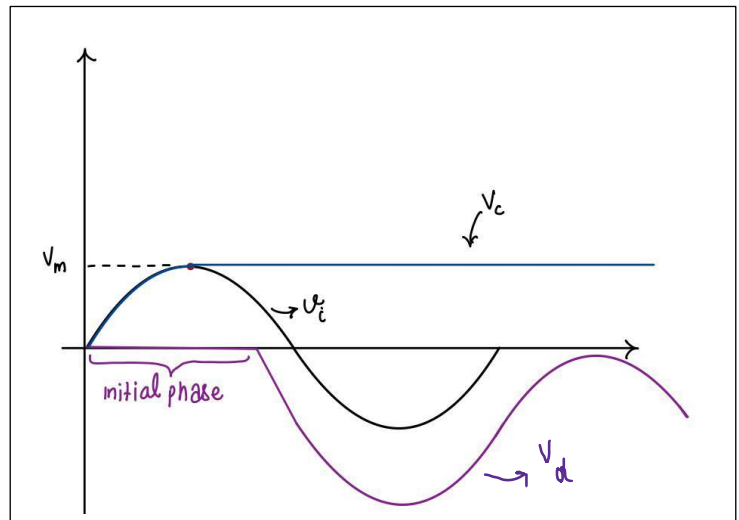
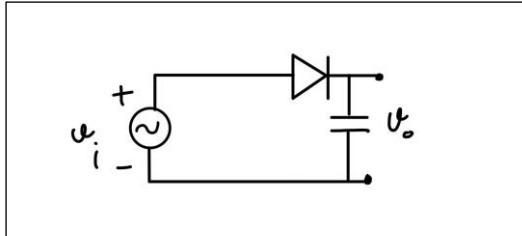


1st lecture

Clamper circuit



The initial phase is when the capacitor is charging,

When its fully charged V_d is

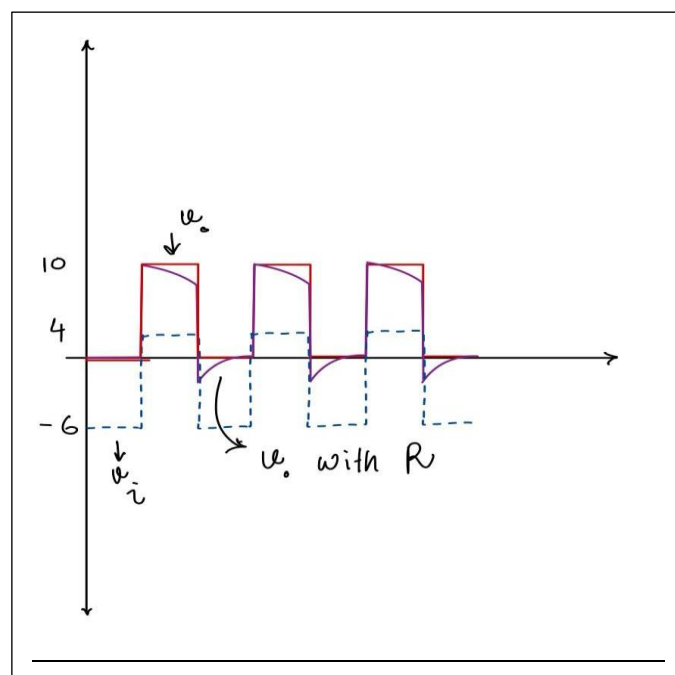
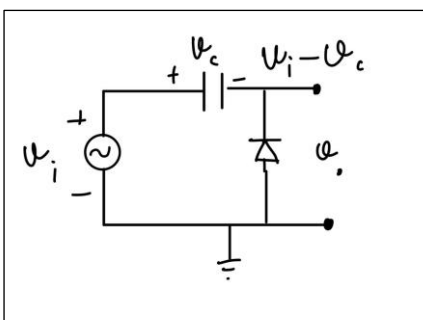
$$v_d = v_i - v_c$$

If the diode is on,

$$v_d > 0$$

$$v_i > v_c$$

Example 1: Find V_o



Assuming C is initially discharged.

Steps:

1. First, we need to find out when the diode will be on (short circuit).

$$v_i - v_c < 0$$

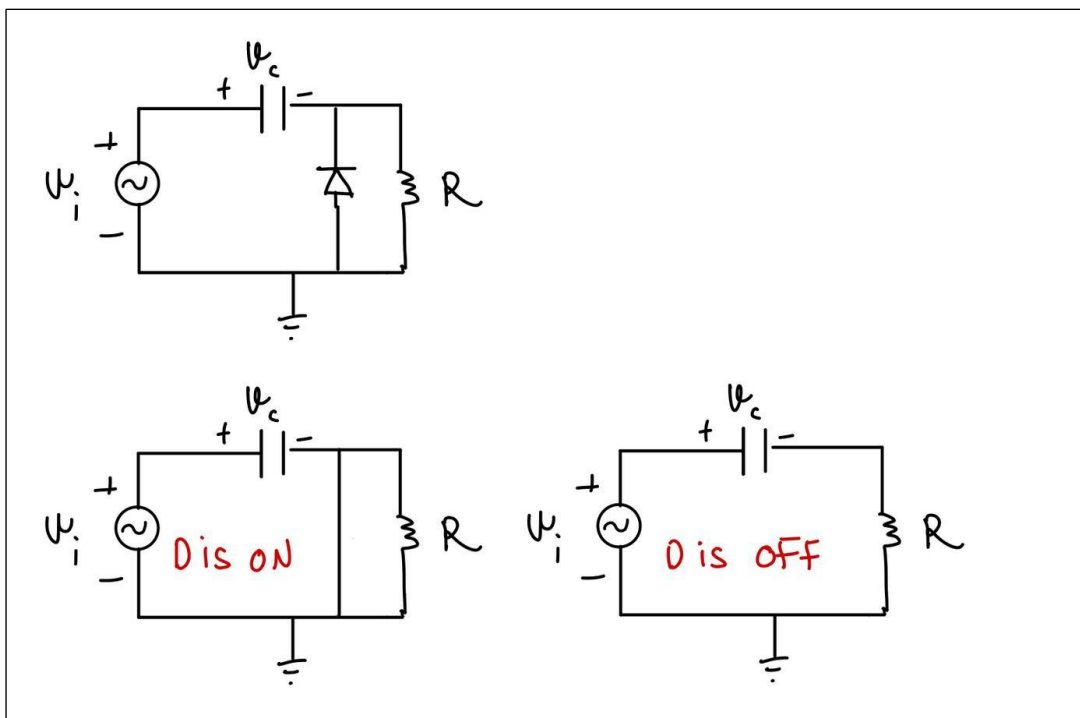
V_c get charged to -6V because V_i applies -6

$$v_c = -6 V$$

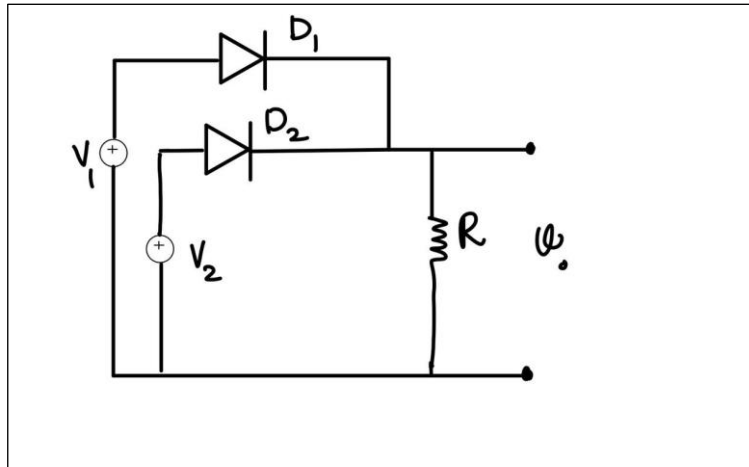
The diode will be off (open circuit) when,

$$v_o = v_i - v_c = v_i + 6$$

**** What happens if we add a resistor in parallel to diode?**



Example 2: Find V_o , given that $V_1 = 5V$, $V_2 = 0V$ and the diodes are Si diodes.



Cases:

1. Both are on

$$v_o = v_1 - 0.7 = 4.3 V$$

$$v_o = v_2 - 0.7 = -0.7 V$$

Therefore, it is not possible.

2. D_1 is on D_2 is off

$$v_o = v_1 - 0.7 = 4.3 V$$

3. D_2 is on D_1 is off:

$$v_o = -0.7 V$$

(but D_1 cannot be off)

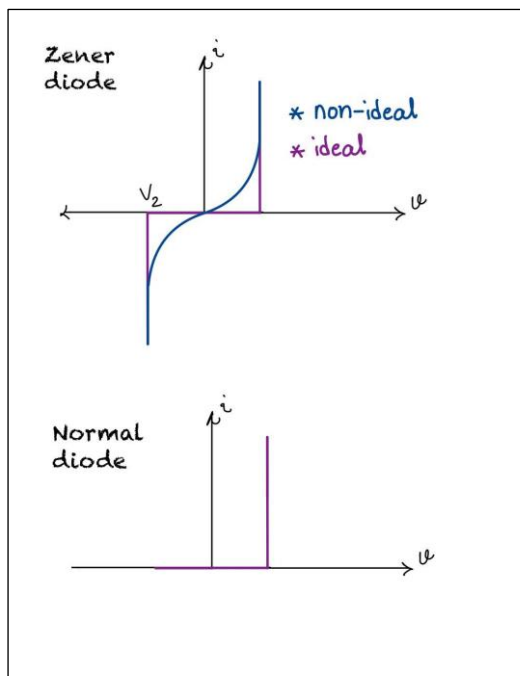
4. Both are off

$$v_o < 0.7 V$$

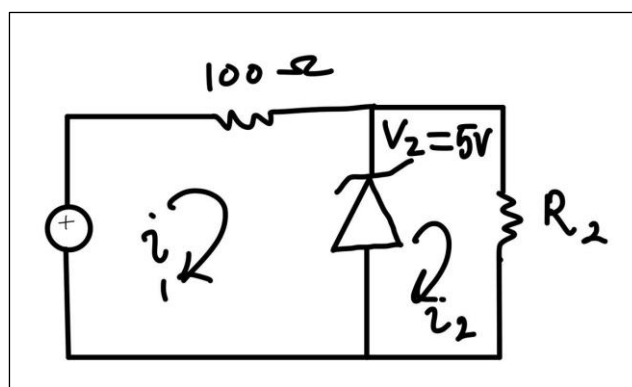
Zener diode

When a Zener diode is reverse-biased and the voltage reaches the Zener (breakdown) voltage, it conducts current while maintaining a nearly constant voltage across it, without being damaged.

The Zener diode can conduct in the reverse direction (allows reverse current)



Example 3: Assuming ideal Zener diode find R_2 , if our purpose is to maintain 5 V across R_2 .



$$\begin{aligned}V_p &= 5 \text{ V} \\10 - 100i_1 - 5 &= 0 \\i_1 &= 0.05 \text{ A}\end{aligned}$$

$$\begin{aligned}5 - R_L i_2 &= 0 \\i_2 &= 5/R_L \\i_{ZD} = i - i_2 &= 0.05 - \frac{5}{R_L}\end{aligned}$$

For Z_D to be on in reverse biased:

$$\begin{aligned}i_{ZD} &> 0 \\0.05 - \frac{5}{R_L} &> 0 \\R_L &> 100 \, \Omega\end{aligned}$$

If Z_D is non-ideal, it implies that minimum 10 mA is needed for reverse breakdown.

$$\begin{aligned}i_{ZD} &> 10 \text{ mA} \\ \text{in this case, } R_L &> 125 \, \Omega\end{aligned}$$

2nd lecture

Operational Amplifiers (OpAmp)

Input voltage is V_+ and V_-

$$V_o = A(V_+ - V_-), \text{ where } A \text{ is the gain}$$

terminal 3 is called the non-inverting terminal

terminal 2 is called the inverting terminal

For ideal OpAmp, $A = \infty$

For non-ideal OpAmp, $A = 10^6$

We can represent OpAmp as a voltage controlled voltage source

R_{in} value – input impedance is very high, it is ∞ for ideal OpAmp

current ≈ 0

Rules for solving:

1. Current entering into input terminal of an OpAmp is 0 (terminals 2 and 3 have 0 current).

Thus it can be used for comparing. If V_+ and V_- but V_+ slightly changes, then V_{out} shows a huge change.

So we can compare V_+ and V_- using V .

Negative feedback

Note: In general, non-inverting amplifier voltage source is connected to non-inverting +ve terminal.

Positive feedback destabilizes the output, negative feedback stabilizes it. We can use the analogy of brakes in a car to understand this.

Connect V_o to V_- (-ve feedback, closed loop)

$$V_o = k(V_+ - V_-)$$

$$V_- = V_o$$

$$V_o = k(V_+ - V_-)$$

Grp 8

$$\frac{V_o}{V_+} = \frac{k}{k+1} = \frac{1}{\ln\left(\frac{1}{k}\right)}$$

As $k \rightarrow \infty$, $V_o \approx V_+$

$V_- \approx V_+$, unity gain buffer

Rule 2 (for negative feedback only)

$$V_+ = V_-$$

Non-inverting amplifier(negative feedback)

$$V_+ = V_i$$

$$V_+ = V_-$$

Nodal analysis,

$$(V_i/R_1) - ((V_i - V_o)/R_2) = 0$$

$$(R_2/R_1) = (V_o/V_i) - 1$$

$$\text{Gain, } A = (R_2 + R_1)/R_1$$

Gain is always > 1

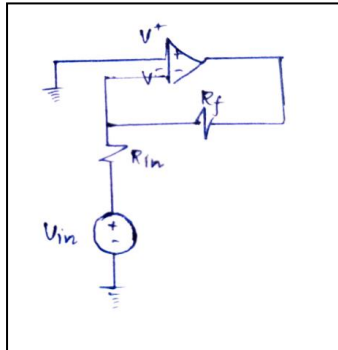
Note:

1. For open loop, gain is ∞ .
2. For negative feedback closed loop, gain is finite, hence it is more practical.
3. By changing R, we can change gain

****Gain=1 when,**

$R_1 = 0$ or $R_2 = \infty$ or when both of the above are true. The circuit changes,
 R_1 becomes open circuit.

Inverting amplifier



- Nodal equation: $\frac{V_0 - 0}{R_f} + \frac{V_{in} - 0}{R_{in}} = 0$

- Expression:

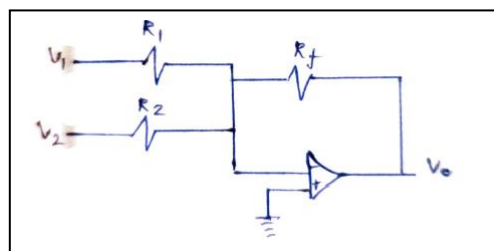
$$\frac{V_0}{R_f} = -\frac{V_{in}}{R_{in}}$$

- Rearranged:

$$\frac{V_0}{V_{in}} = -\frac{R_f}{R_{in}}$$

We can invert the voltage by applying $+V$ and using this inverting circuit. To get unity: make $R_{in} = R_f$, which gives gain = -1.

1. Summer circuit



Nodal analysis:

$$\frac{0 - V_2}{R_2} + \frac{0 - V_1}{R_1} + \frac{0 - V_0}{R_f} = 0$$

$$V_0 = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

If $R_1 = R_2 = R_f = 1\text{k}\Omega$:

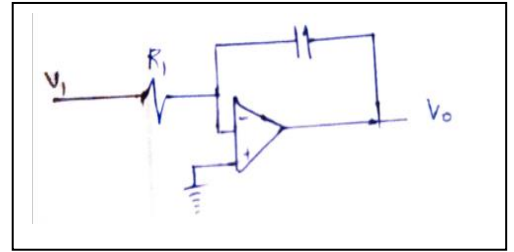
$$V_0 = -V_1 - V_2$$

2. Integrator circuit

Circuit Equation: $\frac{0-V_1}{R_1} + C \left(-\frac{dV_o}{dt} \right) = 0$

$$\frac{dV_o}{dt} = -\frac{V_1}{R_1}$$

Rearranged: $V_o = -\frac{1}{RC} \int V_1 dt$



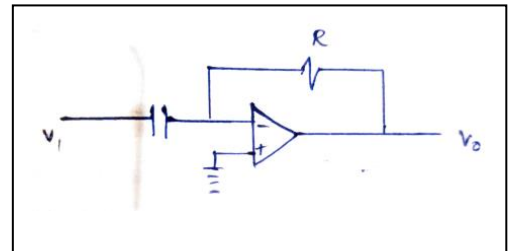
3. Differentiator circuit

Circuit Equation: $\frac{0-V_o}{R} + C \left(-\frac{dV_1}{dt} \right) = 0$

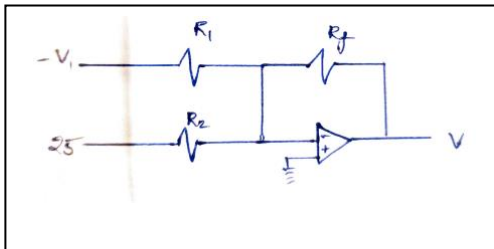
$$-\frac{V_o}{R} = C \frac{dV_1}{dt}$$

Rearranged:

$$V_o = -RC \frac{dV_1}{dt}$$



Example 1: Implement $V = 2V_1 - 25$



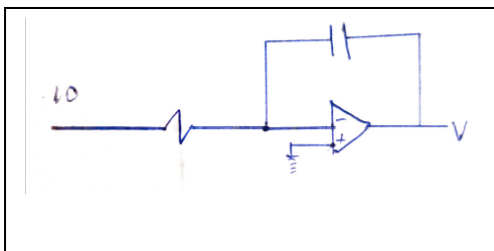
- Circuit equation:
$$V = -(-V_1) \frac{R_f}{R_1} - 25 \frac{R_f}{R_2}$$
$$= V_1 \frac{R_f}{R_1} - 25 \frac{R_f}{R_2}$$

- To satisfy the above equation we need :

$$\frac{R_f}{R_1} = 2 \quad \text{and} \quad \frac{R_f}{R_2} = 1$$

Add inverter to get $-V_1$

Example 2: Implement $\frac{dV_o(t)}{dt} = 10$ (output)



- Circuit Equation:

$$C \frac{dV(t)}{dt} = \frac{10}{R_1}$$

$$\frac{dV(t)}{dt} = \frac{10}{RC}$$

$$RC = 1$$

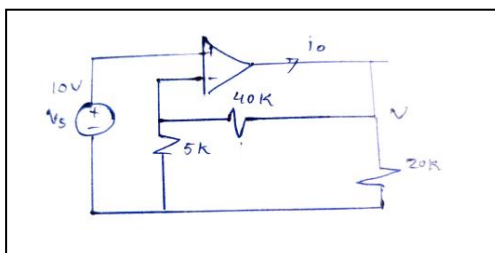
$$R = 10M\Omega$$

$$C = 1\mu F$$

- Add Inverter to get

$$10V \rightarrow -10V$$

Example 3: $i_0 = ?$



- By rule 2 of OpAmp: $V_s = V^-$

- Nodal:

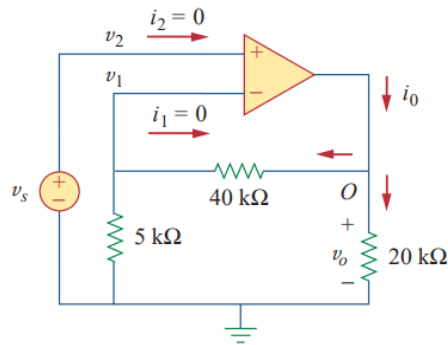
$$\frac{10 - V_0}{40K} + \frac{10 - 0}{5K} = 0$$

- Substitute $V_0 = 90 \text{ Volts}$

$$i_0 = \frac{V_0}{20K} + \frac{V_0 - 10}{40K}$$

$$i_0 = \frac{90}{20K} + \frac{90 - 10}{40K}$$

$$i_0 = 6.5 \text{ mA}$$

Example Questions !**(1) Operational amplifiers (Ideal)**1. Find i_o and V_o/V_s Given $V_s = 1V$ 

Solution:

- Two characteristics of an ideal opamp is
 - Current into both input terminals are zero
 - Voltage difference across input terminals are zero.
- From the diagram, $V_s = V_2$.
- Since $i_1 = 0$, $40\text{ k}\Omega$ and $5\text{ k}\Omega$ are in series.
- V_1 is the voltage across $5\text{ k}\Omega$. We use voltage division to derive V_o/V_s .
- Using nodal analysis at node O, we can find current i_o .

Calculations:

$$V_1 = \frac{5}{40 + 5} V_o = V_o/9$$

We know $V_1 = V_2 = V_s$

$$V_s = V_o/9$$

$$V_o/V_s = 9$$

At node O

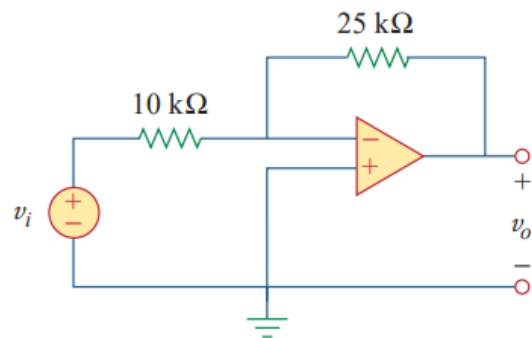
$$i_o = \frac{V_o}{40 + 5} + \frac{V_o}{20} \text{ mA}$$

Substituting $V_o = 9V$ in the above equation, we arrive at

$$i_o = 0.65 \text{ mA}$$

(2) Inverting amplifier

2. If v_i 0.5 V, calculate: (a) the output voltage v_o , and (b) the current in the 10-k resistor.



Solution:

- a. For inverting amplifiers, The gain is given by,

$$\frac{V_o}{V_i} = \frac{-R_f}{R_I}$$

- b. Input the values to obtain output voltage.
 c. To obtain i , since potential difference across the amplifier is zero, node connecting 10K and 25K is at zero potential.
 d. Using $V = iR$ to obtain i .

Calculations:

$$V_o = V_i (-R_f/R_I)$$

$$V_o = 0.5 (-25/10) = -1.25 V$$

The current through 10 k resistor is

$$i = V_i - 0/10 \text{ mA}$$

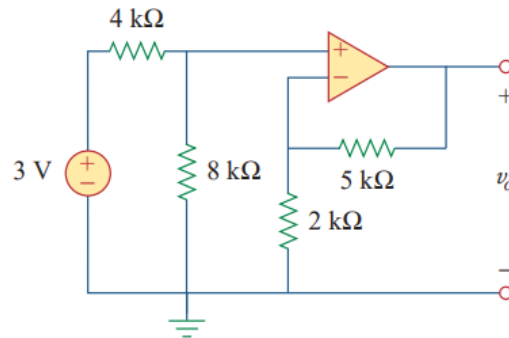
$$i = 0.05 \text{ mA}$$

For this question, you can refer to Fundamentals of electric circuits (4th edition) , Example 5.3, Pg 182, Chapter 5

(3) Non-inverting amplifier

3. Calculate v_o in the circuit given.

Solution:



- Apply voltage divider rule across 8 ohm resistor. (We can recognize it is non inverting since resistance feedback and R_{in} are connected across positive terminal of amplifier)
- Use voltage divider concept in reverse to obtain V_o . (Taken from one of the properties: Voltage difference across input terminals is zero.)
So $V_1 = V_2$
- Another way to approach this would be to use the default formula for non-inverting amplifiers, i.e,

$$\frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_{in}}$$

Calculations:

Using Voltage divider rule,

$$V_1 = \frac{8}{4 + 8} * 3 = 2V$$

$$V_1 = \frac{2}{2 + 5} * V_o$$

$$V_o = 7V \text{ (First method)}$$

$$V_{in} = 2V \text{ (from voltage divider)}$$

$$\frac{V_o}{2} = 1 + \frac{5}{2} = 2 * 3.5 = 7V \text{ (second method)}$$

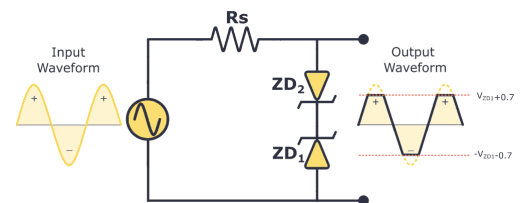
Applications!

A Zener diode is a special type of diode designed to operate in the reverse breakdown region (Zener breakdown) at a specific voltage. It maintains a nearly constant voltage across its terminals when reverse-biased.

Overvoltage Protection

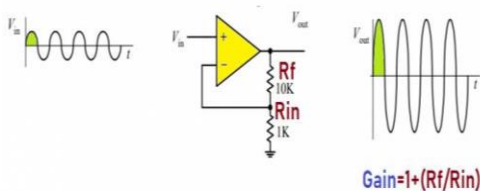
Voltage Regulator

Clipping Circuits



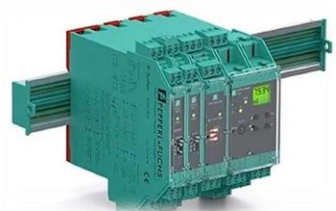
Non- Inverting Amplifier

An op-amp configuration where the input signal is applied to the non-inverting (+) terminal and feedback is applied to the inverting (-) terminal. The output is in phase with the input.



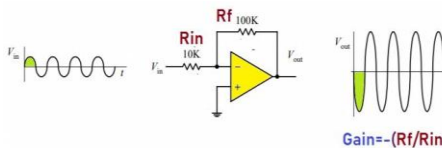
Precision Voltage Reference

Sensor signal conditioning



Inverting Amplifier

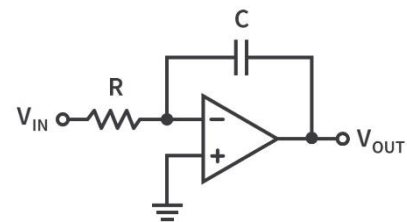
An op-amp configuration where the input signal is applied to the inverting (-) terminal, and feedback is applied from the output to the same terminal. The output is 180° out of phase with the input.



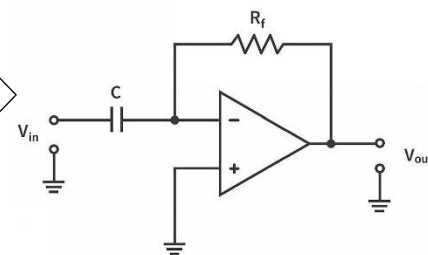
Audio Mixer (summing amplifier)



Integrator Circuits

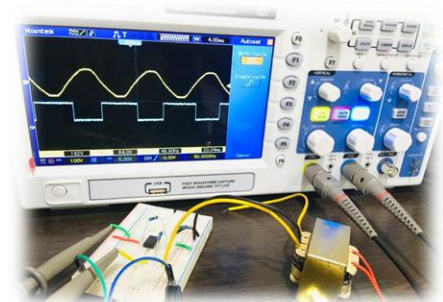


Differentiator Circuit

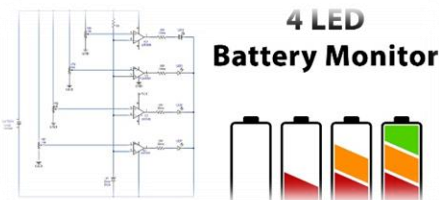


A comparator is an op-amp used in open-loop mode to compare two input voltages. It outputs either a high or low voltage depending on which input is larger.

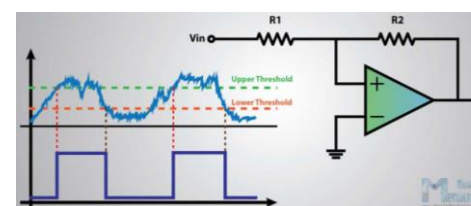
Zero crossing Detector



Level Detection



Schmitt Trigger Circuit (noise immunity)



Contribution statement:

Mahra: compiled all the lecture notes, drew the graphs

Soumyya : 3rd lecture and applications

Tanisha: Example questions and circuit diagrams

Swara: lectures 1 and 2